Algorithm 1 Forward–Backward Sweep Method

Require: Initial state x_0, final time T, control bounds $u_i^{\max}$, weights w_i, tolerance ε, maximum iterations $N_{\max}$

Discretize $[0, T]$ into grid points $\{t_k\}_{k=0}^{n}$

Compute baseline solution by solving the state system with $u_1(t) = u_2(t) = u_3(t) = 0$

Initialize controls:

$$u_1^{(0)}(t_k) = u_2^{(0)}(t_k) = u_3^{(0)}(t_k) = 0$$

for $m = 0, 1, \ldots, N_{\max}$ **do**

  Solve the state equations forward in time:

$$\dot{x} = f(x, u_1^{(m)}, u_2^{(m)}, u_3^{(m)}), \qquad x(0) = x_0$$

  Solve the adjoint equations backward in time:

$$\dot{\lambda} = -\frac{\partial H}{\partial x}, \qquad \lambda(T) = 0$$

  for each grid point t_k **do**

    Update controls using the optimality conditions

$$\tilde{u}_1(t_k) = \min\left\{ u_1^{\max}, \max\left(0, \frac{\lambda_1(t_k) A_\beta(t_k)}{2w_3}\right)\right\},$$

$$\tilde{u}_2(t_k) = \min\left\{ u_2^{\max}, \max\left(0, \frac{\lambda_2(t_k) Ca(t_k)}{2w_4}\right)\right\},$$

$$\tilde{u}_3(t_k) = \min\left\{ u_3^{\max}, \max\left(0, \frac{\lambda_3(t_k) \tau(t_k)}{2w_5}\right)\right\}$$

  end for

  Enforce

$$\tilde{u}_1(0) = \tilde{u}_2(0) = \tilde{u}_3(0) = 0$$

  Apply relaxation:

$$u_i^{(m+1)} = \frac{1}{2}\tilde{u}_i + \frac{1}{2}u_i^{(m)}, \qquad i = 1, 2, 3$$

  Compute relative error

$$E = \max_{i=1,2,3} \frac{\|u_i^{(m+1)} - u_i^{(m)}\|}{\|u_i^{(m)}\| + 10^{-10}}$$

  if $E < \varepsilon$ **then**

    break

  end if

end for

Solve the state and adjoint systems one final time using the converged controls.

return optimal state trajectory $x^*(t)$, adjoint trajectory $\lambda^*(t)$, and controls $(u_1^*(t), u_2^*(t), u_3^*(t))$